



May/June 2026

MODULE CODE: COMP212
MODULE TITLE: Distributed Systems Mock Exam

TIME ALLOWED: TWO Hours

	Additional Information
Calculators Allowed?	No
Student permitted materials	Writing materials
Provided materials	Question paper 1x MCQ answer sheet

Instructions for examination:

- Prepare your answers in the question paper, selecting one option for each question
- When ready, transfer your answers to the MCQ answer sheet provided, shading in the appropriate box for each question

At the end of the examination:

- Please keep the MCQ answer sheet separate from any answer booklets
- **Leave all exam materials, including this question paper, in the examination room**

1. Which layer lies between the physical and network layers in modern distributed systems?
 - ☐ A. Middleware
 - ☐ B. Presentation
 - ☐ C. Session
 - ☒ D. Data link
 - ☐ E. None of the above
2. Which of the following layers in the ISO OSI layered organisation of distributed systems is responsible for the transmission of bits over the network cables?
 - ☐ A. Data link
 - ☐ B. Application
 - ☐ C. Network
 - ☒ D. Physical
 - ☐ E. None of the above
3. Which of the following best describes TCP/IP?
 - ☒ A. A combination of a transport layer and a network layer protocol
 - ☐ B. A transport layer protocol
 - ☐ C. A network layer protocol
 - ☐ D. The de facto error correction suite of the Internet
 - ☐ E. A combination of a transport layer and a data link layer protocol
4. Which of the following best describes a socket?
 - ☒ A. One endpoint of a two-way communication link between two programs running on the network
 - ☐ B. A UDP endpoint
 - ☐ C. Another name for a port
 - ☐ D. A server process that connects to a client
 - ☐ E. A thread on the server's side
5. Which of the following is **not** an example of a distributed system?
 - ☐ A. The Internet
 - ☒ B. A single multiprocessor workstation
 - ☐ C. A banking application
 - ☐ D. The University of Liverpool computer network
 - ☐ E. All of the above are examples of distributed systems

6. Which of the following correctly describes a difference between asynchronous and synchronous communication?
- ☐ A. In asynchronous the client is always blocking and waiting for the server's response
 - ☐ B. In synchronous all clients have access to a global clock and can explicitly synchronise
 - ☐ C. In asynchronous both the client and the server need to be online
 - ☐ D. In synchronous the client can continue to work while the server is processing its request
 - ☐ E. In asynchronous the client can continue to work while the server is processing its request
7. What is the main functionality offered by DNS to clients' web browsers?
- ☐ A. Returning the mail host for a domain
 - ☐ B. Caching
 - ☐ C. Returning the type of hardware and OS of a server
 - ☐ D. Resolving names to IP addresses
 - ☐ E. Resolving IP addresses to names
8. Another question related to systems.
- ☐ A. A possible answer.
 - ☐ B. A possible answer.
 - ☐ C. A possible answer.
 - ☐ D. A possible answer.
 - ☐ E. A possible answer.
9. Another question related to systems.
- ☐ A. A possible answer.
 - ☐ B. A possible answer.
 - ☐ C. A possible answer.
 - ☐ D. A possible answer.
 - ☐ E. A possible answer.
10. Another question related to systems.
- ☐ A. A possible answer.
 - ☐ B. A possible answer.
 - ☐ C. A possible answer.
 - ☐ D. A possible answer.
 - ☐ E. A possible answer.

11. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

12. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

13. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

14. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

15. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

16. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

17. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

18. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

19. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

20. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

21. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

22. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

23. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

24. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

25. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

26. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

27. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

28. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

29. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

30. Another question related to systems.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

31. Which of the following is true for the FloodSet algorithm?

- ☒ A. It solves the agreement problem in complete networks of n processors when at most f processors may crash (i.e., stop) during any execution, given that the algorithm knows f in advance
- ☐ B. It solves the agreement problem in complete networks of n processors when at most f processors may crash (i.e., stop) during any execution, without requiring knowledge of f
- ☐ C. It solves the agreement problem in general networks (i.e., any network) of n processors when at most f processors may crash (i.e., stop) during any execution, given that the algorithm knows f in advance
- ☐ D. All of the above
- ☐ E. None of the above

32. Consider a distributed system consisting of three computers X , Y , and Z . When X 's clock is showing 2:00 the time shown respectively in Y 's and Z 's clocks are 1:55 and 2:20. Assuming X to be the master for this distributed system, the Berkeley algorithm:

- ☐ A. Synchronises the clocks to $X = Y = Z = 2:20$
- ☐ B. Synchronises the clocks to $X = Y = Z = 1:55$
- ☐ C. Synchronises the clocks to $X = Y = Z = 2:00$
- ☒ D. Synchronises the clocks to $X = Y = Z = 2:05$
- ☐ E. Cannot be applied in this case

33. Which of the following best describes the typical complexity measures of the performance of distributed algorithms?

- ☐ A. Time complexity and transparency complexity
- ☐ B. Energy consumption and CPU usage
- ☐ C. Time complexity, communication complexity, and space complexity
- ☐ D. Time complexity only
- ☐ E. Communication complexity only

34. In the context of distributed algorithms what is the purpose of a proof of correctness?

- ☐ A. To establish that a given problem has been correctly formulated
- ☐ B. To establish the communication complexity of the algorithm
- ☐ C. To establish the running time of the algorithm
- ☐ D. To formally convince that a given problem is impossible to solve
- ☐ E. To formally convince that an algorithm does as expected

35. What is the communication complexity (in total number of messages transmitted) of broadcast from a given root u_0 over a given spanning tree T of a network G of n processors and m edges?
- ☒ A. $n - 1$
 - ☐ B. $n + 1$
 - ☐ C. m
 - ☐ D. n^2
 - ☐ E. $O(\log n)$
36. Which of the following best describes the distributed **broadcast** problem from a given processor, in which a spanning tree of the network is given?
- ☐ A. There is a network $G = (V, E)$ and a distinguished root processor u_0 . There is also a spanning tree $T = (V, E')$, where E' is a subset of E , on top of G , and only u_0 knows T . u_0 must consult with the other processors over T in order to **elect** itself a unique leader.
 - ☐ B. There is a network $G = (V, E)$ and a distinguished root processor u_0 . The processors additionally know in a distributed way a spanning tree $T = (V, E')$, where E' is a subset of E . u_0 has some information it wishes to send to all processors.
 - ☐ C. There is a network $G = (V, E)$ and a distinguished root processor u_0 . There is also a spanning tree $T = (V, E')$, where E' is a subset of E , on top of G , and **only u_0 knows T** . u_0 has some information it wishes to send to all processors.
 - ☐ D. All of the above
 - ☐ E. None of the above
37. Consider a distributed system coming with the guarantee that at most 1032 crash failures can happen, for any network of 100 processors in this system. What is the worst-case time complexity of the FloodSet algorithm when executed on a network of 100 processors in this system?
- ☐ A. 1031 seconds
 - ☐ B. 1032 CPU cycles
 - ☐ C. 1032 rounds
 - ☒ D. **1033 rounds**
 - ☐ E. 1031 rounds
38. The coordinated attack problem:
- ☐ A. Represents agreement under reliable communication
 - ☐ B. Represents validity under unreliable communication
 - ☐ C. Represents agreement under unreliable communication
 - ☐ D. Represents agreement under unreliable processors
 - ☐ E. None of the above

39. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

40. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

41. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

42. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

43. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

44. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

45. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

46. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

47. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

48. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

49. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.

50. Another question related to algorithms.

- ☐ **A.** A possible answer.
- ☐ **B.** A possible answer.
- ☐ **C.** A possible answer.
- ☐ **D.** A possible answer.
- ☐ **E.** A possible answer.